

课后习题 (2) - 词法分析 1.

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1. 解: (1) 证明: $L(A|A) = L(A) \cup L(A) = L(A) \therefore A|A = A$

(2) 证明: $L(A^*) = L(A)^* = A^0 \cup A^1 \cup \dots \cup A^n = A^*$

其中 $L((A^*)^*) = L(A^*)^* = (A^*)^0 \cup (A^*)^1 \cup \dots \cup (A^*)^n = A^0 \cup A^1 \cup \dots \cup A^n = A^*$

$\therefore L(A^*) = L((A^*)^*)$ $L(A^*)$ 为任意个 A 组合的串 $L((A^*)^*)$ 是任意个 A^* 组合的串
其中 A^* 是任意个 A 组成的串, 因此 $L((A^*)^*)$ 即为所有由 A 组成的串

$\therefore L(A^*) = L((A^*)^*) \therefore A^* = (A^*)^*$

(3) 证明: $L(\epsilon|AA^*) = L(\epsilon) \cup L(AA^*) = \{\epsilon\} \cup (L(A)L(A^*)) = \{\epsilon\} \cup \{A\} \cup L(A^*)$

其中 $L(A^*) = \{\epsilon, A, AA, \dots, \text{任意个 } A \text{ 的串}\} \therefore \{\epsilon\} \cup \{A\} \cup L(A^*) = L(A^*)$

$\therefore L(\epsilon|AA^*) = L(A^*) \therefore A^* = \epsilon|AA^*$

(4) 证明: $L((AB)^*A) = L((AB)^*)L(A) = L(AB)^*L(A) = \{(AB)^0 \cup (AB)^1 \cup \dots \cup$

$(AB)^n\} \{A\} = (AB)^0 \{A\} \cup (AB)^1 A \cup (AB)^2 A \cup \dots \cup (AB)^n A$

$= A(BA)^0 \cup A(BA)^1 \cup A(BA)^2 \cup \dots \cup A(BA)^n = L(A(BA)^*)$

$\therefore (AB)^*A = A(BA)^*$

2. 解: (1) $(0|1)^*01$

(2) $(015) | (1|2|3|4|5|6|7|8|9)(011|2|3|4|5|6|7|8|9)^*(015)$

(3) 要使 0 或 1 出现奇数个, 我们可以使用 $(AA)^*A$ 的方式

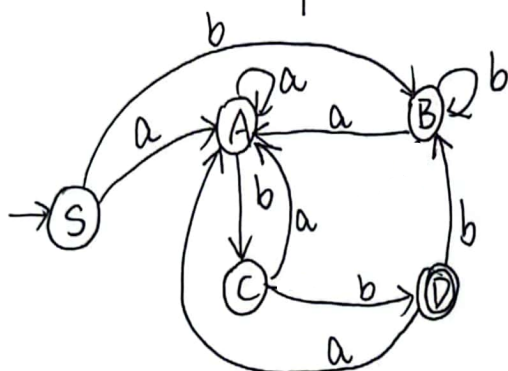
因此. $((0^*|0^*)(0^*|0^*))^*(0^*|0^*) | ((1^*|01^*)(1^*|01^*))^*(1^*|01^*)$

(4) $a^*b^*c^*d^*e^*f^*g^*h^*i^*j^*k^*l^*m^*n^*o^*p^*q^*r^*s^*t^*u^*v^*$
 $w^*x^*y^*z^*$

3. 解.

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	I_a	I_b
$\{0, 1, 2, 4, 7\} (S)$	$\{3, 8, 6, 7, 1, 2, 4\} (A)$	$\{5, 6, 7, 1, 2, 4\} (B)$
$\{3, 8, 6, 7, 1, 2, 4\} (A)$	$\{3, 8, 6, 7, 1, 2, 4\} (A)$	$\{5, 9, 6, 7, 1, 2, 4\} (C)$
$\{5, 6, 7, 1, 2, 4\} (B)$	$\{3, 8, 6, 7, 1, 2, 4\} (A)$	$\{5, 6, 7, 1, 2, 4\} (B)$
$\{5, 9, 6, 7, 1, 2, 4\} (C)$	$\{3, 8, 6, 7, 1, 2, 4\} (A)$	$\{5, 10, 6, 7, 1, 2, 4\} (D)$
$\{5, 10, 6, 7, 1, 2, 4\} (D)$	$\{3, 8, 6, 7, 1, 2, 4\} (A)$	$\{5, 6, 7, 1, 2, 4\} (B)$



4. 解.

$\{S_4\}$ $\{S_0, S_1, S_2, S_3\}$

$a \downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 S_1 S_1 S_1 S_1
 $b \downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 S_2 S_3 S_2 S_4

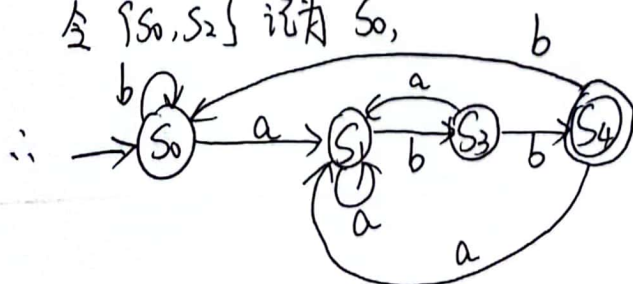
$\{S_4\}$ $\{S_3\}$ $\{S_0, S_1, S_2\}$

$a \downarrow$ $\downarrow$ $\downarrow$
 S_1 S_1 S_1
 $b \downarrow$ $\downarrow$ $\downarrow$
 S_2 S_3 S_2

$\{S_4\}$ $\{S_3\}$ $\{S_1\}$ $\{S_0, S_2\}$

由此可得划分如左所示

令 $\{S_0, S_2\}$ 记为 S_0 ,

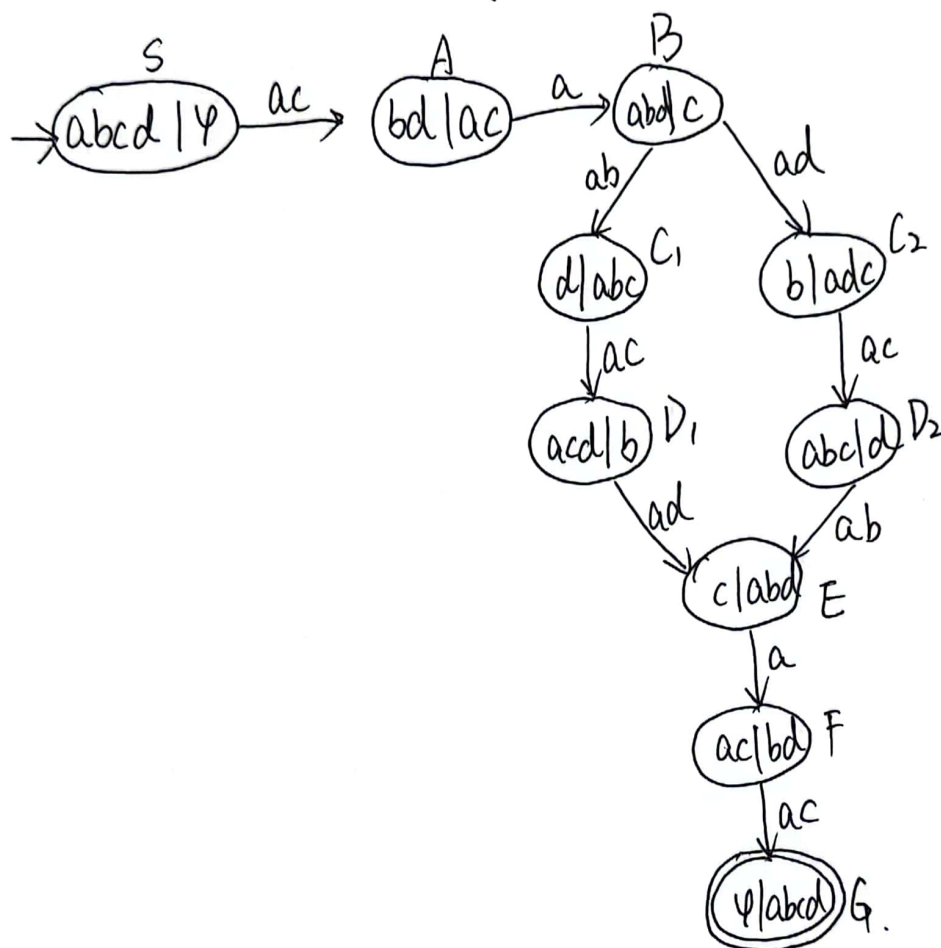


5.解: 有可能. 设人为a, 狼为b, 羊为c, 白菜为d,

用"|"表示河, 其两侧表示两岸, 由题意可知, bc, cd不可同时出现

其中初始状态为 $abcd|\varnothing$, 终态为 $\varnothing|abcd$.

DFA M的图示: 其中 $M = \{s, A, B, C_1, C_2, D_1, D_2, E, F, G\}, \{ac, a, ab, ad\}, f, s \{G\}$



整体流程就是人先带走羊, 留下狼和白菜,

人回来, 带走狼(或白菜),

人带羊回来, 留下狼(或白菜)

人带走白菜(或狼)

人回来, 带走羊, 即完成.